

Transformer Loading Analysis - TGI Use Case Notes

[UC26-01] - Awesense Energy Transition Platform Use Case



Overview

This manual is intended to:

- Provide instructions to create the selected use case in True Grid Intelligence (TGI).

For more details about Transformer Loading Analysis and how it can be analyzed using Awesense's Energy Transition Platform, please refer to the [UC26-01 - Transformer Loading Analysis.pdf](#) document.

TGI Use Case Notes

This document shows the implementation process of mentioned use case in the TGI visualization interface along with a description of how to create such a use case on your data.

The transformer loading analysis can be found directly in the TGI for a transformer element if that transformer has measured data¹. The easiest way is to find a transformer via the TGI Explorer in the main menu (select from the main menu - top hamburger icon) and then select the transformer from the map or table. Clicking on a transformer will open a sidebar on the left side with additional information, and at the bottom of this sidebar is the "Phase Balance" section where clicking on "View Report" will display this report.

The screenshot shows the True Grid Intelligence (TGI) interface. On the left sidebar, the 'Grid Element' section is active, displaying details for transformer 17AY23-8-9. Below this, the 'Trace' section shows various options like Source, Same Voltage, Connected, etc. The 'Phase Balance' section is visible, and a 'View Report' button is highlighted. The 'Grid Element Details' section shows information like label, name, and rating. The main area displays a map with transformer locations and a table of transformer data.

ID	Circuit	Element Type	Phases	Latitude	Longitude	Is Producer	Is Consumer
10001	North Central Zone	Transformer	C	39.254601464	-82.533343845	No	No
10005	North Central Zone	Transformer	A	38.596587474	-82.431991662	No	No
10007	North Central Zone	Transformer	B	39.180105578	-82.055933487	No	No
1001	North Central Zone	Transformer	B	38.723686737	-82.339261635	No	No
10010	North Central Zone	Transformer	A	38.8827073	-82.692737045	No	No
10012	North Central Zone	Transformer	A	38.816414542	-82.324232864	No	No
10013	North Central Zone	Transformer	A	38.834501148	-82.518109731	No	No
10014	North Central Zone	Transformer	B	38.769297398	-82.764101722	No	No
100145	North Central Zone	Transformer	C	38.600791303	-82.415557345	No	No
10017	North Central Zone	Transformer	A	38.668167494	-82.411648443	No	No

¹ Transformer measured data can be originating directly from measurement device installed on given transformer or TGI has option to aggregate downstream consumption and infer the load flow on the transformer this way.

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The basic interface of the Transformer Technical Analysis report offers information about the analyzed transformer and its attributes. Further, the interface offer quick information about the percentage of the time the transformer is under or over loaded.

Transformer Technical Analysis

Transformer Overview

Element ID:	transformer_92
commission_date:	1995-01-01
installation_type:	station
is_operation_status:	Yes
label:	tr_92
latitude:	49.26829145
load_loss:	1117
longitude:	-123.10273
maintenance_type:	dry
name:	tr_92
no_load_loss:	194
ownership:	utility
primary_voltage:	12500
rating_kva:	15
secondary_voltage:	400
voltage_level:	MV/LV

First data point - 18 May, 2023 12:56 PM

PF Override: PF

Percentage of Time Overloaded

35.8%

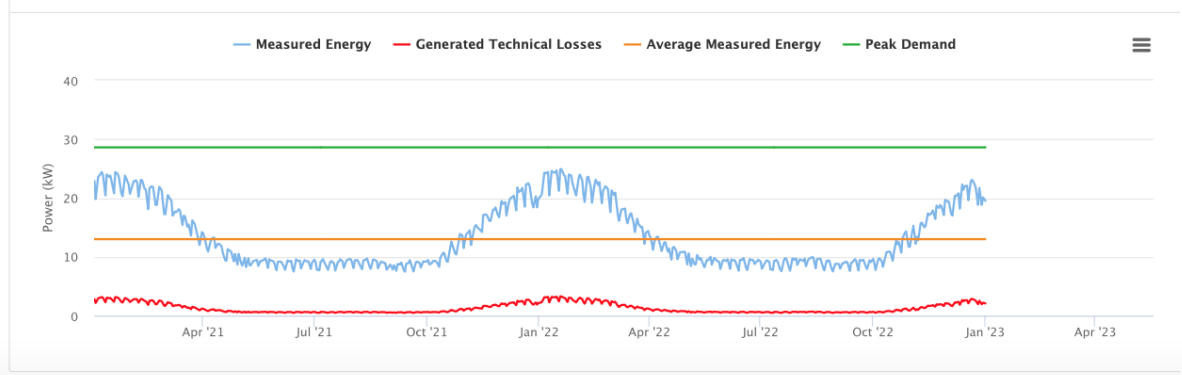


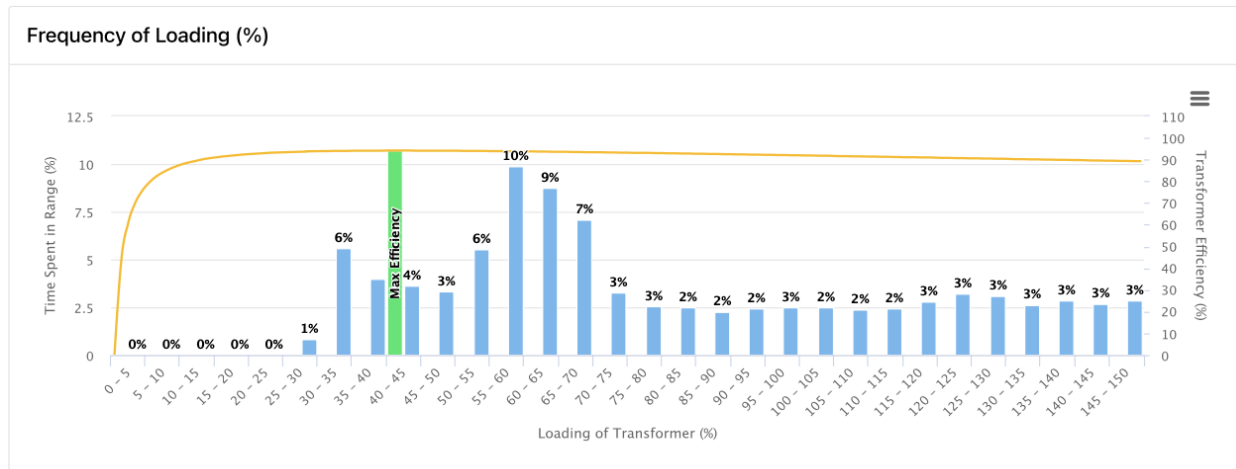
Percentage of Time Underloaded

0.845%



Transformer Performance





The interface also offers the the possibility to choose different time periods.

The Transformer Performance graph shows the transformer load pattern over the selected period and also shows other indicators such as Peak Demand, Generated Technical Losses and Average Measured Energy.

The last graph shows a histogram of the transformer loading. Here you can clearly see how the transformer was loaded most often. The transformer loading efficiency (yellow) is also shown along with the maximum efficiency loading ratio (green column).

General Note

This description was created for TGI version 3.10.0. For future versions of TGI, the process might be slightly different. Please contact tools@awesense.com for more information.